AI Optical Biosensors for Non-invasive Neurotransmitter Detection

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1 Introduction

Neurotransmitters are the chemical messengers inside the brain that do communication between the neurons. They also follow certain cycle , understanding this cycle is essential for making biosensor that will detect the chemicals .

2 Recent advancements

In recent years advancements have been done in the neurotransmitters like integration of ai. machine learning and deep learning which helps to seperate overlapping signals , analyze the complex signals given by the biosensors and improve accuracy .

AI also helps real time monitoring and prediction . After that optical biosensors came . these sensors use light based technique like fiber optics . When neurotransmitter interact with the sensor surface , they make changes in the light properties like intensity , wavelength etc. this changes are converted into measurable signals .

After that came nanotechnology which played a key role in improving the sensor performance as the nanoparticles increase the surface areas and enhances the interaction with target molecules which increases the sensitivity and selectivity . they also helps in developing small and portable sensors .

Then comes the modern research which focuses on non-invasive techniques rather than invasive implants .(non invasive are the sensors on body surface). Nanotransmitters can be detected using the bodyfluids like blood , saliva or sweat.

3 Limitations

With advancments limitations also exist . Sensors perform well in laboratory conditions but fails in real world . Long tern stability and biocompatibility remains challenge . AI modles require large and high quility dataset also often lack transparency . there in also lack of standardization .